Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc2`; Mathlib at commit `1b1ccf4b`; physlib (`master`) and TauCeti (commit `e7e7f52`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Inconsistent convergence in Hilbert space

Files.

- File: `briefs/InconsistentConvergence/InconsistentConvergence.lean`
- Library: `InconsistentConvergence`
- Imports: `Mathlib.Analysis.Normed.Lp.lpSpace`
- Second file, for the oscillator: module `InconsistentConvergence.Oscillator`, in the subdirectory `InconsistentConvergence/`. It is separate from the main file, so that the comparator lane, which certifies the main file, is unchanged. Its imports:
  - `InconsistentConvergence`;
  - `ForMathlib.Analysis.Distribution.SchwartzSpace.HermiteReal`;
  - `TauCeti.Analysis.SpecialFunctions.Hermite.Function.Oscillator`.

  The `ForMathlib` file identifies $\mathcal{S}(\mathbb{R})$, with its topology, with the rapidly decreasing Hermite series. It transfers the multi-index Hermite expansions on $\mathbb{R}^1$, in the same directory, to $\mathbb{R}$.

- Third file, for the operators: module `InconsistentConvergence.NumberOperator`, in the same subdirectory and separate from the main file for the same reason. Its imports:
  - `InconsistentConvergence`;
  - `ForMathlib.Analysis.InnerProductSpace.l2Diagonal`;
  - `ForMathlib.LinearAlgebra.LinearPMap`.

  The `ForMathlib` files provide the diagonal operators on ℓ^2, self-adjoint on their maximal domains, and the composition of unbounded operators on its natural domain.
- Fourth file, for the Schwartz sequence space, in the same subdirectory and separate from the main file for the same reason: module `InconsistentConvergence.SchwartzSequences`. Its imports:
  - `InconsistentConvergence.Oscillator`;
  - `ForMathlib.Analysis.Normed.Lp.SchwartzSeq`.

  The `ForMathlib` file constructs $S(\mathbb{N})$ as a topological vector space.
- Fifth file, for the observables on $S(\mathbb{N})$, in the same subdirectory and separate from the main file for the same reason: module `InconsistentConvergence.SchwartzObservables`. Its imports:
  - `InconsistentConvergence.NumberOperator`;
  - `InconsistentConvergence.SchwartzSequences`.

  The construction of continuous linear maps of $S(\mathbb{N})$ from weighted bounds, and the continuity of their expectations, are in `ForMathlib.Analysis.Normed.Lp.SchwartzSeq`.
- Sixth file, for unbounded observables in general, in the same subdirectory and kept apart from the main file for the same reason: `InconsistentConvergence.UnboundedObservables`. Its imports:
  - `InconsistentConvergence.NumberOperator`;
  - `ForMathlib.Analysis.InnerProductSpace.UnboundedForm`.

  The `ForMathlib` file proves the general statements, for densely defined operators on a complex inner product space.

Setting. $H = \ell^2(\mathbb{N})$ (`lp (fun _ : ℕ => ℂ) 2`), with $|n\rangle = \delta_n$ the oscillator eigenstates and

$$|e^j\rangle = \sqrt{\tfrac{j-1}{j}}\,|0\rangle + \sqrt{\tfrac{1}{j}}\,|j\rangle\,.$$

In the oscillator file, U is the unitary $\ell^2(\mathbb{N}) \to L^2(\mathbb{R})$ of the Hermite basis (`TauCeti.hermiteHilbertBasis`), with $U\,|n\rangle = \psi_n$, the n-th Hermite function ($m = \omega = \hbar = 1$). The oscillator file describes $S(\mathbb{N})$ by the seminorms $\sup_n (n+1)^K |x_n|$, $K \in \mathbb{N}$; the fourth file constructs it as a topological vector space, with the brief's seminorms.

Results.

Brief	Lean
eq. (2): $\|e^j - \|0\rangle\|^2 = \left(\sqrt{(j-1)/j} - 1\right)^2 + 1/j$	`norm_sub_ground_sq`
$e^j \to \|0\rangle$ in ℓ^2	`tendsto_e`
eq. (3): $\sum_n n^2 \|e^j_n\|^2 = j \to \infty$	`secondMoment_e`, `tendsto_secondMoment_e`
the $\|j\rangle$ alone have no limit in ℓ^2	`not_cauchySeq_level`: $\|\,\|j\rangle - \|k\rangle\,\|^2 = 2$ for $j \neq k$
eq. (4): $\langle e^j \| N \| e^j \rangle = 1$ for $j \geq 1$	`numberExp_e`
eq. (5), the inconsistency	`not_continuousAt_numberExp`: $\langle N \rangle$ is not continuous at $\|0\rangle$ in ℓ^2

Results: the oscillator.

Brief	Lean
$U\|n\rangle = \psi_n$	`hermiteUnitary_level`; the coordinates of $U^{-1}g$ are $\langle \psi_n \| g \rangle$ (`hermiteUnitary_symm_apply`)
$\|n\rangle$ is the oscillator eigenstate with n quanta	`oscillator_level`: $-\psi_n''/2 + x^2\psi_n/2 = (n + \frac{1}{2})\psi_n$
$S(\mathbb{N})$ is the Schwartz space	`exists_schwartz_iff`: $Ux \in \mathcal{S}(\mathbb{R})$ iff $\sup_n (n+1)^K \|x_n\| < \infty$ for every K
and has its topology	`tendsto_schwartz_iff`: $f_j \to g$ in $\mathcal{S}(\mathbb{R})$ iff $U^{-1}f_j \to U^{-1}g$ in every seminorm of $S(\mathbb{N})$
the e^j are Schwartz states	`exists_schwartz_e`: $Ue^j \in \mathcal{S}(\mathbb{R})$
$e^j \to \|0\rangle$, but not in $S(\mathbb{N})$	`tendsto_hermiteUnitary_e`: $Ue^j \to \psi_0$ in $L^2(\mathbb{R})$; `not_tendsto_schwartz_e`: Ue^j has no limit in $\mathcal{S}(\mathbb{R})$

Results: the number operator. In the operator file, a and $a^\dagger$ act by $(ax)_n = \sqrt{n+1}\, x_{n+1}$, $(a^\dagger y)_0 = 0$ and $(a^\dagger y)_{n+1} = \sqrt{n+1}\, y_n$, and N by $(Nx)_n = n\, x_n$, each on its maximal domain: $D(a) = \{x \mid (\sqrt{n}\, x_n)_n \in \ell^2\}$ and $D(N) = \{x \mid (n\, x_n)_n \in \ell^2\}$.

Brief	Lean
$a\|n+1\rangle = \sqrt{n+1}\ \|n\rangle$, $a\|0\rangle = 0$, $a^\dagger\|n\rangle = \sqrt{n+1}\ \|n+1\rangle$	`annihilation_level_succ`, `annihilation_level_zero`, `creation_level`
$a^\dagger$ is the adjoint of a	`adjoint_annihilation`
N is self-adjoint, with $N\|n\rangle = n\|n\rangle$	`isSelfAdjoint_numberOp`, `numberOp_level`; domain: `mem_domain_numberOp`
$N = a^\dagger a$	`numberOp_eq_adjoint_compMax`, with the composition on $\{x \in D(a) \mid ax \in D(a^\dagger)\}$
$\langle x \| N \| x \rangle = \sum_n n\,\|x_n\|^2$	`inner_numberOp`, on $D(N)$; `numberExp_eq_norm_sq`: $\sum_n n\,\|x_n\|^2 = \|ax\|^2$ on $D(a)$
eq. (4) for the operator	`inner_numberOp_e`: $e^j \in D(N)$ and $\langle e^j \| N \| e^j \rangle = 1$ for $j \geq 1$

Results: the Schwartz sequence space. In the fourth file, $S(\mathbb{N})$ is the type `SchwartzSeq` of sequences with $((n+1)^k x_n)_n \in \ell^2$ for every k, with the topology of the seminorms $\|((n+1)^k x_n)_n\|$.

Brief	Lean
$S(\mathbb{N}) \subset \ell^2$, with the topology of its seminorms	`SchwartzSeq.withSeminorms`; the inclusion into ℓ^2 is continuous (`SchwartzSeq.continuous_toLp`)
Prop. 1: $e^j \to a$ in ℓ^2, but e^j does not converge in $S(\mathbb{N})$	`tendsto_toLp_eS`, `not_tendsto_eS`: $\|((n+1)\, e^j_n)_n\| \geq \sqrt{j}$ (`sqrt_le_seminorm_eS`)
each b^j is in $S(\mathbb{N})$, but the sequence has no limit in $S(\mathbb{N})$	`levelS`, `not_tendsto_levelS`
$S(\mathbb{N})$ is the Schwartz space of the oscillator	`hermiteSchwartzEquiv`: $S(\mathbb{N}) \cong \mathcal{S}(\mathbb{R})$ as topological vector spaces, given by U (`toLp_hermiteSchwartzEquiv`), with $\|n\rangle \mapsto \psi_n$ (`toLp_hermiteSchwartzEquiv_levelS`)

Results: observables on $S(\mathbb{N})$. In the fifth file, a, $a^\dagger$ and N are restricted to $S(\mathbb{N})$, which lies in their domains. The bounds $(n+1)^k|(ax)_n| \leq p_{k+1}(x)$, $(n+1)^k|(a^\dagger x)_n| \leq 2^k p_{k+1}(x)$ and $(n+1)^k|(Nx)_n| \leq p_{k+1}(x)$, with $p_k(x) = \|((n+1)^k x_n)_n\|$, make them continuous linear maps $S(\mathbb{N}) \to S(\mathbb{N})$ (`SchwartzSeq.mkCLM`).

Brief	Lean
a, $a^\dagger$, N act on $S(\mathbb{N})$	`annihilationS`, `creationS`, `numberOpS`, agreeing with the operators of the fourth file (`toLp_numberOpS`); $N = a^\dagger a$ (`numberOpS_eq_comp`)
the Schwartz topology does not allow this kind of convergence: $\langle N \rangle$	`continuous_inner_numberOpS`, `continuous_numberExp`: $\langle x\|N\|x\rangle$ is continuous on $S(\mathbb{N})$
Prop. 1, from the expectation	`not_tendsto_eS_of_numberExp`: a limit of e^j in $S(\mathbb{N})$ would be $\|0\rangle$, with $\langle N \rangle = 0 \neq 1$
the same for all moments	`continuous_inner_of_mem_adjoin`: $\langle x\|T\|x\rangle$ is continuous on $S(\mathbb{N})$ for every polynomial T in a and $a^\dagger$, among them $\hat{x} = (a + a^\dagger)/\sqrt{2}$, $\hat{p} = i(a^\dagger - a)/\sqrt{2}$ and $N + \frac{1}{2}$ (`positionS`, `momentumS`, `hamiltonianS`)

Results: unbounded observables. The general statements are in `ForMathlib`, for an operator A with dense domain $D(A)$ in a complex inner product space, unbounded in the sense that no C has $\|Ax\| \leq C\|x\|$ on $D(A)$. The instances are in the sixth file, with the position $\hat{x} = (a + a^\dagger)/\sqrt{2}$ of the oscillator on $D(a) \cap D(a^\dagger) = D(a^\dagger)$.

Brief	Lean
the domain of an unbounded self-adjoint operator cannot be the whole space	`LinearPMap.domain_ne_top_of_isFormalAdjoint`: Hellinger–Toeplitz, from Mathlib's `LinearMap.IsSymmetric.continuous`
a continuous linear operator is bounded	Mathlib; in this project also `HilbertUnphysicality.not_continuous_of_unbounded`
an unbounded operator is not continuous in its own domain: small state changes, big changes of the average	`LinearPMap.not_continuousAt_inner`: $x \mapsto \langle x\|A\|x\rangle$ is discontinuous at every point of $D(A)$; `LinearPMap.not_continuousAt_inner_div`: so is $\langle x\|A\|x\rangle / \|x\|^2$ at every $x \neq 0$
$\langle N \rangle$	`exists_lt_norm_numberOp`, `not_continuousAt_inner_numberOp`, `not_continuousAt_numberExp_domain`: discontinuous at every point of $D(N)$
the average position is not defined for all states	`exists_lt_norm_positionOp`: $\|(\hat{x}\,\|n\rangle)_{n+1}\| = \sqrt{(n+1)/2}$; `isFormalAdjoint_positionOp`, `domain_positionOp_ne_top`: $D(\hat{x}) \neq \ell^2$
and is not continuous in its domain	`not_continuousAt_inner_positionOp`, `not_continuousAt_inner_div_positionOp`

Differences from the brief.

- Indexing. Proposition 1 indexes sequences from 1, with the ground term at $i = 1$. The formalization indexes from 0, matching the oscillator of Section 3. The divergence is the same; with the brief's indexing the sum is $(j-1)/j + j$ instead of j.
- No limit in $S(\mathbb{N})$. In the main file, what is proved is that the seminorm $\sum_n n^2|x_n|^2$ is unbounded along e^j, which is the brief's argument. That e^j has no limit in the topology of $S(\mathbb{N})$ is proved in the fourth file.
- Seminorms of $S(\mathbb{N})$. The brief's seminorms $\sum_i |i^k x_i|^2$, with $i \geq 1$, are the squares of $\|((n+1)^k x_n)_n\|$ with indices from 0. The oscillator file uses the seminorms $\sup_n (n+1)^K |x_n|$, which define the same space and the same topology (`SchwartzSeq.mem_submodule_iff`, `SchwartzSeq.tendsto_iff`).
- In the main file, $\langle N \rangle$ is defined on all of ℓ^2 as the weighted sum $\sum_n n\,|x_n|^2$, which has a default value outside the domain of N. The discontinuity is still meaningful, because every e^j is finitely supported. The operator file constructs $N = a^\dagger a$ and shows that the weighted sum is $\langle x|N|x\rangle$ on $D(N)$.
- All moments. The brief says that the Schwartz topology "keeps track of all moments". The formalization reads this as the continuity on $S(\mathbb{N})$ of the expectations of the polynomials in a and $a^\dagger$. The proof does not use the polynomial form: the expectation of every continuous linear map $S(\mathbb{N}) \to S(\mathbb{N})$ is continuous (`SchwartzSeq.continuous_inner_apply`), and the polynomials are such maps because a and $a^\dagger$ are.
- Unbounded operators. The brief argues that a continuous operator is bounded because it extends from a dense domain to the whole space. The formalization uses directly that a continuous linear map is bounded. The discontinuity in the domain is proved for the expectation

$\langle x|A|x\rangle$, which is the brief's "average", and holds for every densely defined unbounded operator, symmetric or not, since over $\mathbb{C}$ the expectation determines $\langle x|A|y\rangle$ by polarization. The position is that of the oscillator in ℓ^2; under the Hermite unitary U it is multiplication by x in $L^2(\mathbb{R})$, which is not formalized here.

Problems found in the brief. No mathematical problems. Minor: the distance is written $d(x,y) = \sum_i (x_i - y_i)^2$, which is the squared distance and, for complex entries, should read $\sum_i |x_i - y_i|^2$.